

SCHOLAR: Engineering an Auditable Local-First Assistant for Scientific Documents

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Abstract

Private and unpublished scientific PDFs motivate document assistants that can run locally and expose the source behind an answer. We present SCHOLAR, a local-first assistant with source-scoped evidence identities, an auditable answer trace, and bounded post-generation repair. We evaluate practical design choices on 100 retrospective retrieval questions and 50 paired generation cases on an 18 GB workstation. BM25 retrieves a gold page first for 84% of questions; adding dense retrieval reaches 85%, whereas a modality-ranking heuristic reduces first-page hits to 59%. Repair raises automatic reference-unit overlap F1 from 0.159 to 0.224 but lowers cited-page alignment from 0.670 to 0.618 on the 20 answerable cases with page labels. In 20 cross-paper controls, the pipeline’s abstained status falls from 13 to 5 cases after repair. A separate LLM-based evaluation of 150 legacy answers records 59 correct, 33 partly correct, 20 incorrect, and 38 not judgeable. These are retrospective model judgments, not human ratings. They show that inspectable provenance does not establish support, and that retrieval, repair, and refusal policies need separate validation.

1 Introduction

Scientific papers mix prose, equations, tables, and figures, yet researchers often need to search private or unpublished PDFs without sending them to a hosted service. Retrieval-augmented generation (RAG) can expose source material at inference time (Lewis et al., 2020), but a deployed assistant must also distinguish a real citation from a supporting citation, a lexical match from entailment, and a refusal in final prose from an internal ABSTAINED status.

We study these distinctions in SCHOLAR, a local-first PDF question-answering system. It ingests a document into a versioned evidence bundle, retrieves source-scoped pages and regions, answers

through a local model, and records every citation, lexical support decision, edit, and terminal outcome. The intended use is researcher-assisted reading on a consumer workstation; this is an implemented prototype, not evidence of production deployment or user benefit.

The contributions are: (1) an operational design that separates acquisition-enabled setup from strict-local analysis and keeps evidence identity stable through the answer path; (2) trace-audited evaluations of retrieval fusion, deterministic post-generation repair, and post-hoc LLM-based answer assessment; and (3) adverse findings that qualify common reliability claims. In our retrospective corpus, BM25 plus dense retrieval offers little first-page gain over BM25, the tested modality heuristic harms ranking, and repair changes both citation alignment and pipeline abstention in undesirable directions. The LLM judge also identifies many incomplete answers. We report these as diagnostics, not as proof of correctness or user benefit.

The paper moves from prior scientific-document systems to ScholAR’s trust boundaries and repair mechanism, then defines the evaluation before reporting results and the operational lessons they support. This order keeps implementation claims, measured outcomes, and deployment implications distinct.

2 Related Work

Scientific-document question answering. QASPER grounds information-seeking questions in full research papers (Dasigi et al., 2021); M3SciQA and SPIQA extend this setting to multiple papers and visual evidence (Li et al., 2024; Banerjee et al., 2024). Recent Industry Track work grounds table–text questions in enterprise annual reports and validates generated cases with domain experts (Wang and Balloccu, 2026). These benchmarks define important evaluation targets.

ScholAR instead studies the system contracts needed to preserve inspectable evidence identity from private PDF ingestion through a local answer and its audit trace.

Visual document retrieval. ColPali showed that late-interaction embeddings over page images can retrieve visually rich documents without reducing every page to OCR text (Faysse et al., 2025). Large-scale document extraction studies likewise find that image-only and OCR-assisted pipelines have different operational failure modes (Shen et al., 2026). ScholAR therefore keeps lexical, dense-text, crop-image, and full-page channels distinct in the trace. It also separates source pixels from model-generated visual descriptions so that a model observation cannot silently become its own supporting evidence.

Attribution and selective generation. ALCE evaluates correctness and citation quality for retrieval-grounded long-form answers (Gao et al., 2023), while FActScore decomposes long-form output into atomic factual claims (Min et al., 2023). SciFact provides evidence-centered scientific claim verification (Wadden et al., 2020). Self-RAG learns when to retrieve and critique generations (Asai et al., 2024), and recent deployment-oriented RAG work adapts retrieval depth to estimated evidence sufficiency (Verma et al., 2026). ScholAR does not claim a stronger learned verifier. Its contribution is an origin-aware execution contract: model citations, application-added mappings, support decisions, bounded edits, abstentions, and failures remain separate, versioned records that production and evaluation consume identically.

These distinctions motivate the system design that follows: before evaluating answer quality, we first make document identity, evidence origin, and every answer transformation inspectable.

3 System Design and Trust Boundaries

Canonical document state. Ingestion constructs a versioned bundle of source bytes, pages, structural blocks, chunks, visual regions, metadata, and content hashes. Before publication it checks source identity, pagination, evidence identifiers, image checksums, and relational constraints. Staging, per-document locking, atomic publication, and rollback prevent partially processed bundles from becoming queryable. A citation identity includes its source document, so “Figure 2” in one paper cannot re-

solve to an identically named figure in another.

Locality and retrieval. Acquisition-enabled setup may fetch documents or model assets. Strict-local analysis permits prepared uploads, caches, and loopback model services but rejects external network requests before name resolution. Query retrieval combines BM25 (Robertson and Zaragoza, 2009) and dense text rankings through reciprocal-rank fusion; crop and page-image channels provide inspection candidates for visual questions. Structural-reference expansion may bring a figure or table named by retrieved text into context. A rule-based modality heuristic can additionally boost selected visual or table candidates, but our ablation finds that this version harms first-page ranking. Retrieved pixels and model-generated visual descriptions remain different artifact types; a model description is never treated as independent source evidence.

One answer path and trace. Production and evaluation call the same versioned pipeline. It records the exact prompt evidence, raw and final completions, model and prompt identities, candidate ranks, citation origins (model-emitted versus application-added or remapped), lexical check outcomes, repair edits, abstention reason, terminal error, and stage timings. The interface exposes complete stages rather than token streaming. Figure 1 shows the trust boundaries, not a claim that every supported-looking output is factually correct.

With those provenance boundaries fixed, the next section specifies the deliberately narrow checks and edits applied after generation.

4 Claim Checking and Bounded Repair

The post-processor splits generated text into factual spans while retaining character offsets and treating citation markers as separate spans. It resolves cited evidence IDs to source pages and records whether a citation came from the model, mapping, or repair. A deterministic token-overlap score compares each claim with its cited source text; numeric mismatch and threshold rules assign operational labels. These labels are *lexical routing decisions*, not validated entailment or contradiction judgments. Pixel-only claims remain explicitly unverified. A negative-control test of the lexical support rule failed, so we do not use its labels as the paper’s correctness endpoint.

Repair uses a bounded action set: keep a span,

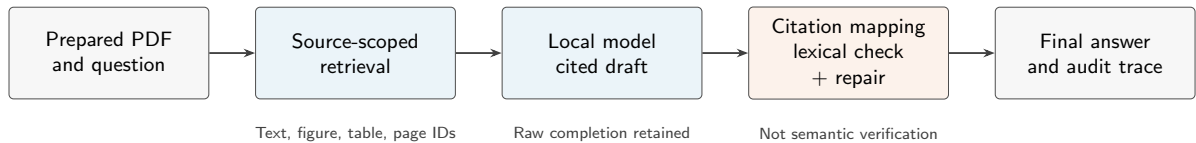


Figure 1: **Auditable local answer path.** A prepared PDF and question enter source-scoped retrieval; selected text or pixels go to a local model; citation mapping and bounded lexical repair produce the final answer and trace. The lexical check is not an entailment judge.

narrow a partial clause, attempt one citation remap, delete a span, or abstain if no claim remains. It does not generate replacement facts. A second pass records the resulting spans and edits, but reusing the same lexical rule is not independent confirmation. The paired study isolates this post-processing intervention by applying both settings to the same raw completion and prompt. It asks whether the rule changes observable overlap, page alignment, and status; whether humans find the repaired answer better is unmeasured.

This separation between an unchanged raw answer and a bounded post-processor determines the comparisons and denominators in the evaluation design.

5 Evaluation Design

Retrospective corpus and tasks. The retrieval study uses 100 curated questions over 25 previously prepared scientific papers, with page-level gold evidence. Thirty answerable questions over 19 papers (10 each for mathematical, textual, and visual queries) test generation with and without post-generation repair. Twenty cross-paper swaps over 16 target papers probe responses when the requested fact was selected from a different source. A phrase-absence check passed for every swap, but semantic absence was not independently judged. These samples are not a paper-disjoint test set and should not be read as population estimates.

Comparisons and measurements. Retrieval compares keyword overlap, BM25, dense text, BM25+dense reciprocal-rank fusion (RRF), and the same fusion with a modality heuristic. A hit requires at least one gold page among distinct retrieved pages; the reported Hit@5 is not fractional recall over all gold pages. We also report mean reciprocal rank (MRR) and page nDCG@5. Answer comparisons share the exact raw completion and prompt within every paired case; only post-processing differs. Automatic token F1, reference-

unit overlap F1, key-term coverage, and cited-page alignment are transparent proxies, not human factuality or citation-support judgments. Pipeline SUCCESS and ABSTAINED describe execution outcomes, not the meaning of the final prose.

Post-hoc LLM-based answer evaluation. We also audit 150 answers from an earlier 200-question run over 10 papers, excluding the disjoint 50-case packet reserved for a planned human study. Exact question matching and a nonempty reference make 113 cases eligible for correctness review; 37 fail this gate but remain usable for grounding and citation checks against supplied excerpts. The batch workflow records correctness, completeness, evidence grounding, citation quality, and one failure type per case. Its retained rows identify the evaluator only as `manual-assistant-proxy`, although the planning file names a specific LLM, so we call this an LLM-based evaluation but do not make a named-model claim. We likewise exclude the visual-grounding distribution: its completed rows use a `partly-supported` label absent from the frozen visual rubric. The remaining counts are post-hoc model judgments, not independent human annotations or a paper-disjoint test.

Human evaluation plan and current status. A private packet assigns the same 50 questions to five independent evaluators, for 250 planned ratings. The development-overlapping sample contains five questions from each of 10 papers; evaluators score correctness, completeness, grounding, citation quality, and utility on five-point scales, verify the supplied reference, and label every evidence item. At the September 21, 2026 manuscript freeze, the study database records zero submitted ratings across zero cases, so no human result is reported. After complete returns, the analysis slot will contain question-clustered bootstrap intervals, all 10 evaluator-pair agreement comparisons per dimension, reference-verification counts, evidence labels, and category and difficulty breakdowns.

Retriever	Hit@1	Hit@5	MRR	nDCG@5
Keyword	.71	.91	.796	.757
BM25	.84	.97	.892	.855
Dense text	.76	.95	.838	.796
BM25+dense RRF	.85	.98	.905	.853
RRF+modality	.59	.95	.734	.714

Table 1: **The modality heuristic harms first-page retrieval.** Retrospective page-level results on 100 questions/25 papers. Hit@k means any gold page at k, not fractional recall.

Execution and uncertainty. The paired generation study used the same pinned Qwen 3.5 9B model (Qwen Team, 2026), quantized as Q4_K_M, on an 18 GB M3 Pro in strict-local mode. Requests ran in terminal batches of at most five under a 50% free-memory start gate; the service and model were unloaded between batches. The audit verifies all expected condition–case keys, dataset and model identities, identical raw paired completions and prompts, and separate errors and abstentions. A defective five-case control pilot was retained for audit but excluded; the corrected 20-case pairs were rerun. We use 10,000 paired bootstrap resamples clustered by paper for 95% intervals. Latencies are wall-clock observations of these runs, not controlled warm-cache comparisons. The legacy 200-question result file used by the LLM-based evaluation does not retain an equivalent generator digest, so those verdicts are not merged with the paired intervention results. With these scopes fixed, we report retrieval first, then repair and controls, LLM judgments, human-study status, and operational latency.

6 Evaluation Results

BM25+dense improves Hit@1 over BM25 by only .01 (paired 95% CI $[-.054, .078]$), so its incremental benefit is unresolved. Adding the modality heuristic lowers Hit@1 by .26 (CI $[-.388, -.136]$) and MRR by .172 (CI $[-.253, -.092]$) relative to BM25+dense. This negative ablation is consequential: extra visual intent and structural boosts can displace a correct first page even while Hit@5 remains relatively high. It does not isolate which sub-rule caused the loss.

Repair changes the final text in all 30 answerable cases. Its overlap gains are small in absolute terms and do not establish human-judged factual support. The 10 remaining cases name a figure or table as the gold locus but provide no gold page, so treating their unscored alignment as zero would

Automatic diagnostic	Off	On	Δ (95% CI)
Token F1	.047	.067	+.020 [.015, .027]
Ref.-unit F1	.159	.224	+.065 [.042, .091]
Key-term coverage	.759	.742	-.017 [-.038, .000]
Cited-page alignment	.670	.618	-.052 [-.091, -.015]

Table 2: **Repair improves lexical overlap but worsens measured page alignment.** Thirty paired answerable cases; intervals are paper-clustered paired bootstraps. Page alignment is defined for only the 20 cases with a gold page label. All metrics are automatic proxies.

LLM-judge aspect	Recorded distribution
Correctness ($n = 113$)	59 correct; 33 partly correct; 20 incorrect; 1 not judgeable
Grounding ($n = 150$)	112 supported; 26 partly supported; 10 unsupported; 2 not judgeable
Citation quality ($n = 150$)	110 accurate; 28 partly accurate; 10 inaccurate; 2 missing

Table 3: **The LLM judge identifies both answer and evidence failures.** Correctness uses only reference-eligible cases; 37 additional cases failed the reference gate. Counts are post-hoc model labels, not human ratings.

be invalid. The pipeline records 28/30 SUCCESS without repair and 30/30 with repair, but success is not correctness.

The missing *gold page labels* are concentrated entirely in the 10 visual questions: none has a paired page-alignment value, whereas all 10 mathematical and all 10 textual questions do. All 10 visual outputs have page-bearing citations in both conditions; those pages cannot be checked against a gold visual locus with this annotation. Exploratory category-level overlap gains appear in all three groups (Appendix Table 5), but they do not establish visual support.

For the 20 cross-paper controls, ABSTAINED falls from 13/20 without repair to 5/20 with repair (paired change $-.40$, CI $[-.619, -.188]$). Four cases in each condition abstain before generation. Some SUCCESS outputs nevertheless refuse in prose, while others discuss unrelated target-paper content; status is thus not a valid semantic-refusal measure. The observed direction is adverse for a status-based safety story, and no hallucination-reduction claim follows.

Table 3 reports the completed LLM-based evaluation without turning its ordinal labels into a single score. One reference-eligible case was still marked not judgeable because the recorded reason incorrectly called its reference ineligible; we preserve that outcome rather than recoding it. Across all 150 cases, the most frequent non-`none` failure la-

Measured duration	p50 (s)	p95 (s)
End-to-end	27.34	149.54
Retrieval stage	0.52	112.85
Generation stage	26.59	37.48

Table 4: **Generation dominates the median; first-use retrieval contributes to the tail.** One local workstation, 100 valid requests, uncontrolled cache state. Stage quantiles need not add to end-to-end quantiles.

bel is `missing_fact` (53 cases), followed by `wrong_fact` (14). These labels are mutually exclusive judgments from one LLM, not error rates with independent adjudication. The visual labels are omitted for the protocol mismatch described in Section 5.

Human evaluation analysis (pending). At the September 21, 2026 manuscript freeze, the reserved 50-question study has 0 of 250 planned ratings and 0 of 50 cases with any rating. Accordingly, there are no human means, confidence intervals, agreement estimates, reference-verification outcomes, or evidence-support counts to analyze. This paragraph is the explicit insertion point for those planned outputs after all five independently completed files have been validated and imported; until then, Table 3 must not be described as human evaluation.

Table 4 profiles 100 valid requests; end-to-end p90 is 41.61 s. Generation dominates the median, while first-use index construction contributes to the retrieval tail. We observed no terminal errors in the corrected batches. We did not measure peak process memory or controlled cold/warm latency; a batch-order difference cannot be interpreted as a repair speedup.

Taken together, the results distinguish what is inspectable from what is supported: the trace exposes each decision, while the negative ablations and incomplete human evidence constrain the conclusions. The next section turns those constraints into deployment lessons.

7 Operational Lessons

Provenance must be separated from support. Source-scoped IDs make cited pages inspectable and prevent cross-document identifier collisions, but they do not establish that a page supports a claim. The low automatic answer overlap and missing gold-page labels for visual questions illustrate why citation validity, page alignment, and indepen-

dent claim support require separate evaluation.

More components need not improve the system.

The modest BM25+dense difference and the large loss from modality boosting challenge an “add every channel” design. The heuristic should remain disabled in a deployment configuration until a revised rule succeeds on independent documents. Likewise, deterministic repair is easy to audit but cannot be promoted as a safety mechanism from lexical overlap alone: the control statuses and page alignment move in the wrong direction.

Locality and latency are operational contracts.

Strict-local analysis requires a prepared model and document bundle; acquisition and installation are separate network-enabled activities. The observed median request is tens of seconds, with a long first-use retrieval tail. This profile supports feasibility on the named workstation, not interactive responsiveness at scale. A deployment decision requires controlled cache conditions, peak memory, workload throughput, and actual researcher task outcomes.

Evaluation provenance is part of quality assurance.

The LLM-based pass is useful because it exposes missing facts and weak evidence, but its model-identity and rubric mismatches prevent stronger claims. A deployable evaluation service should bind the judge provider, model version, prompt, schema, inputs, and outputs in the same trace and reject out-of-schema updates before aggregation. Completed rows are not enough when their provenance contract is incomplete, and they do not replace the pending human study. Together, these lessons frame the conclusion: auditability is an operational foundation, not a substitute for validated answer quality.

8 Conclusion

SCHOLAR makes local document assistance inspectable by preserving evidence identity and separating raw generation, citation mapping, lexical checks, repair, and terminal status. The retrospective study reveals a near-null dense-retrieval gain, a harmful modality-ranking heuristic, and a repair policy that raises lexical overlap while worsening cited-page alignment and abstained-status frequency. Among 113 reference-eligible cases in the LLM-based evaluation, 59 are labeled correct, 33 partly correct, 20 incorrect, and one not judgeable; the retained evaluator identity is insufficient for a named-model claim. These findings argue

for independent evidence and refusal judgments before treating provenance or repair as reliability guarantees. The prepared human study has not yet received ratings, and a paper-disjoint human evaluation and real-user deployment study remain necessary.

Limitations

The evaluation is retrospective and curated: the 100 retrieval questions and 50 paired generation cases do not form a paper-disjoint, independently sampled test set. The 150-case LLM-evaluation sample is disjoint from a reserved 50-case annotation packet, but both come from the same legacy 200-question, 10-paper corpus. Its source result file lacks the generator digest recorded for the paired study; its retained judge label does not establish the named model in the planning file; and one eligible case has a contradictory not-judgeable reason. We exclude its visual-grounding counts because the completed rows use a category absent from the frozen rubric. No LLM label is an independent human judgment. At the September 21, 2026 freeze, the human study has 250 planned but zero submitted ratings, and is itself development-overlapping rather than paper-disjoint. Automatic lexical overlap can reward copying, miss paraphrases, and overlook wrong numbers or unsupported implications; a failed lexical negative control prevents us from treating verifier labels as semantic support. Cross-paper phrase absence is not proof that a question is semantically unanswerable. Pipeline status is not equivalent to a refusal in prose. Page alignment is observable for only 20 of 30 answerable cases per condition because visual gold loci name figures or tables without page labels. The results use one pinned local model and one 18 GB workstation; memory observations are pre-window free-memory checks rather than peak process measurements, and latency includes uncontrolled cache effects. The prototype has not been tested in a real researcher workflow, and no user benefit, production robustness, throughput, or cross-domain generalization is established. PDF extraction and reading order remain potential error sources. Local analysis does not imply that initial model or document acquisition is offline. The system should not be used for autonomous high-stakes decisions; users must inspect primary evidence.

Ethical Considerations

Local processing can reduce document disclosure, but it does not eliminate privacy risk: PDFs, extracted text, figures, embeddings, traces, prompts, and model outputs remain on the device and require deletion, access-control, and retention practices appropriate to the user’s environment. The interface discloses the active network mode, and the paper distinguishes external acquisition from strict-local analysis. Benchmark documents and model artifacts must be used under their licenses. Human claim annotation and the researcher pilot must not begin before the responsible institution provides the applicable ethics or IRB determination; consent, compensation, de-identification, exclusions, and data retention must then follow the approved process. Citation-like output may create unwarranted trust, so the system exposes source evidence and selective failure and must not market a retrieved page as proof that a statement is true.

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A Reproducibility and Diagnostic Detail

Trace accounting. Each expected (condition, case) pair has exactly one terminal row; interrupted runs resume missing keys and reject duplicates. The retained fields include source, model digest and quantization, prompt hash, raw completion, final text, retrieved evidence identities, citation origins, edit actions, terminal status, and stage timings. Aggregation reads immutable traces rather than regenerating answers. The answerable and cross-paper pairs have identical raw completions and prompt hashes within all 50 paired cases. The five-case control pilot with an adapter classification defect is retained with its checksum and excluded from the corrected 20-case comparison.

Uncertainty and denominators. The retrieval rows cover 100 questions/25 papers and 500 condition outputs. Answerable rows cover 30 cases/19 papers per repair setting; control rows cover 20 cases/16 target papers per setting. All corrected rows have zero terminal errors. Paired intervals resample papers, not individual questions, 10,000 times. The citation-page diagnostic uses the 20 cases whose gold locus specifies a page in both answerable conditions. The remaining 10 gold loci name a figure or table but no page; they are missing labels, not zero-scored failures. Four controls per condition abstain before generation. Consequently, the 13/20 versus 5/20 status comparison cannot be interpreted as a pure post-generation classifier experiment.

Type	<i>n</i>	Ref.-unit F1 off/on	Page labels
Visual	10	.129/.200	0/10
Mathematical	10	.173/.218	10/10
Textual	10	.175/.254	10/10

Table 5: Exploratory subgroup diagnostics. Automatic reference-unit overlap is not factual correctness; gold-page counts mark cases eligible for the page-alignment score. Visual outputs do cite pages, but the gold visual loci do not supply page labels.

Observed failure patterns. The modality heuristic can promote intent-matched visual or table candidates ahead of a gold page. Lexical repair edits all 30 answerable responses and can change citation mapping as well as wording. Some final outputs with pipeline SUCCESS contain refusal prose; others address target-paper material that does not answer the swapped question. These patterns motivated the explicit distinction among status, lexical

overlap, cited-page alignment, and independently judged support. We omit document excerpts and raw completions from this review artifact because source permissions and anonymity have not been cleared.

Artifact scope. The reproducibility package contains source code, schema and protocol definitions, tests, a small synthetic fixture for trace-accounting checks, and instructions to rerun authorized local evaluation. It excludes private PDFs, document databases, raw local completions, user records, model weights, and cached embeddings. The printed numbers are trace-audited retrospective findings; the public fixture validates accounting software, not the empirical scores.

LLM-evaluation accounting. The LLM-evaluation manifest contains 150 unique cases, 15 from each of 10 papers, and is the exact complement of a 50-case human-study packet. It binds the source dataset, legacy result file, and annotation bundle by SHA-256. The public receipt contains 150 judgments and records 113 reference-eligible and 37 reference-ineligible cases. We retain two audit exceptions. First, the judgment rows and public aggregate use the generic model identity `manual-assistant-proxy`, which does not verify the model named in the planning protocol. Second, the later visual pass introduced `partly_supported`, although the frozen visual schema permits only `supported`, `unsupported`, `not applicable`, or `not judgeable`. We therefore use only the non-visual categorical counts in Table 3 and do not present the LLM judge as independently validated.